

Bachelor of Science in Information Technology

Subject in Introduction in Computing 1

**"Analyzing a Decade of Computing Research: Insights
from 2015 to 2024"**

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Submitted To

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January 29, 2025

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INTRODUCTION

Technology has advanced at an extraordinary pace, significantly influencing daily life and work by enhancing efficiency and convenience. Computing has been a key driver in the progression of human civilization. Over the past decades, the field of computing systems has undergone remarkable transformations, profoundly shaping society through groundbreaking developments such as the Internet and the commoditization of computing. Propelled by technological innovation, computer systems have continually evolved and adapted to address diverse societal needs. This evolution has given rise to new paradigms like cloud computing, fog computing, edge computing, and the Internet of Things (IoT), creating fresh economic and creative opportunities. The progression of computing, from mainframes and workstations to the cloud and decentralized architectures like edge computing and IoT, retains core components and characteristics essential to their functionality. As we look ahead, the 2020s are expected to witness similarly rapid and significant advancements. According to Moore's law, semiconductor performance is projected to improve 32-fold over a decade, driving extraordinary innovations in the digital age that will not only transform technology but also profoundly impact society. This study aims to provide an analytical overview of the advancements in computing over the past decades.

RESEARCH METHODOLOGY

The methodology of this study is in **qualitative research methods**, chosen for the ability to explore the complex and multifaceted nature of computing research over the past decades (2015–2024). This approach facilitates a comprehensive thematic analysis of significant topics in the field, enabling the researcher to uncover patterns, insights, and interconnections across various domains. The analysis focuses on nine key areas:

- 1) The evolution and ethics of artificial intelligence (AI),
- 2) breakthroughs in quantum computing,
- 3) the growth and impact of open-source software,
- 4) cybersecurity threats and innovations,
- 5) the role of social media algorithms in modern society,
- 6) advances in computer hardware, computing and environmental sustainability,
- 7) the development of virtual reality (VR) and augmented reality (AR),
- 8) and trends in computing education.

To gather insights, the study synthesizes information from scholarly articles, reports, and case studies. This qualitative approach allows for a deep exploration of the ethical considerations,

societal impacts, and technological advancements within each area. By examining these critical topics, the study provides a nuanced perspective on the evolution of computing and its broader

implications, emphasizing the interconnectedness of technology, society, and ethics.

SUMMARY OF FINDINGS

This study is a qualitative study which was designed to analyse articles about the advancements of computing research in the past decades (2015-2024). An overview of the advancements of computing research in the specific fields.

Evolution of Artificial Intelligence (AI)

Artificial Intelligence (AI), a field focused on replicating human cognitive abilities, began in the 1950s with Turing's Test and the Dartmouth Conference, which marked its formal inception. In the 1980s, AI concentrated on symbolic approaches through expert systems, though these systems struggled to fully mimic expert behavior. The 1990s to 2010s saw AI expand into various fields, driven by mathematical tools and cross-disciplinary collaboration. A significant turning point came in 2006 with Fei-Fei Li's ImageNet, showcasing the power of large datasets for AI advancements. Between 2015 and 2019, AI reached major milestones: DeepRL mastered Atari games, AlphaGo defeated a Go champion, AlphaZero beat the top chess AI, and groundbreaking technologies like Transformers and GPT-2 advanced language tasks. Key achievements in robotics, gaming, and language models continued to push AI boundaries. This rapid evolution is fueled by the openness of AI research, open-source frameworks, and global collaboration via platforms like Kaggle and MOOCs. AI has since been applied across numerous domains, including computer vision, speech recognition, medicine, and gaming, driving widespread adoption and innovation.

Ethics in Artificial Intelligence and Computing

According to Cousera (2024), AI ethics refers to the principles and guidelines that ensure AI technology is developed and used responsibly, emphasizing safety, security, humanity, and environmental considerations. A strong AI code of ethics can include avoiding bias, ensuring privacy of users and their data, and mitigating environmental risks. Codes of ethics in companies and government-led regulatory frameworks are two main ways that AI ethics can be implemented. In 2018, the European Commission formed the High-Level Expert Group on AI (HLEG), which established ethical guidelines for trustworthy AI, focusing on three main components: a) legality (compliance with laws), b) ethical adherence (respecting fundamental principles), c) and robustness (ensuring no unintentional harm). Additionally, four core ethical principles are key to trustworthy AI: a) respect for human autonomy, b) prevention of harm, c) fairness, d) and explainability. These principles align with existing legal frameworks but go beyond them, serving as ethical imperatives for AI practitioners. Several institutions, including Stanford's HAI, The Alan Turing Institute, and others, are dedicated to AI ethics research.

transparency, and accountability in AI.

Breakthroughs in Quantum Computing

Quantum computing is a rapidly advancing field with the potential to transform technology, offering solutions to problems once deemed unsolvable. From 2016 to 2023, significant milestones in research and development, such as quantum supremacy and efforts in scaling and error correction, paved the way for future breakthroughs. Quantum computing's potential applications span drug discovery, climate modeling, and financial optimization, with hybrid quantum-classical models playing a key role in tackling complex problems. However, challenges such as hardware stability and scalability remain, and ongoing research is crucial. The future of quantum computing focuses on practical applications, cloud access, and international collaboration, while addressing ethical issues like encryption and equitable access.

Growth and Impact of Open-Source Software

Open source software has advanced significantly in the past decade, moving from foundational components like operating systems to more complex applications, including desktop software (e.g., OpenOffice, Firefox) and enterprise applications for tasks like content management and customer relationship management. A survey of 512 companies found that 87% were using open source software, with large companies (over \$1 billion in revenue) adopting it widely due to in-house IT support. The primary concerns preventing some companies from using open source include support, intellectual property issues, and lack of knowledge or expertise. However, over half of those not using open source expect to adopt it in the next three years. Open source software is transforming the software industry, with many products incorporating open source components or relying entirely on open source. Its presence in both industry and government has led to substantial cost savings, with large companies saving millions annually. Despite its smaller market share, open source revenue reached \$1.8 billion in 2006, and its influence is growing, often integrated into proprietary products. Research shows exponential growth in open source projects and code, highlighting its increasing commercial importance. These trends suggest a rising role for open source in the software market in the future.

Cybersecurity Threats and Innovations

In an era of rapid technological advancement, financial institutions must balance digital transformation with robust cybersecurity. The integration of AI, blockchain, and cloud computing offers opportunities to enhance operational efficiency and customer experiences, but also introduces new cybersecurity risks. AI can aid in real-time threat detection and automated security processes, but it requires continuous refinement and ethical standards. Blockchain

provides decentralized security for transactions but demands rigorous audits and secure smart contracts. Cloud computing offers scalability but presents vulnerabilities, necessitating strong access controls, encryption, and regular assessments. A culture of cybersecurity awareness among employees is essential, as human error is often a significant vulnerability. Financial institutions must adopt a holistic approach, integrating technological innovation with strong security frameworks. By investing in cybersecurity and fostering awareness, organizations can protect sensitive data, adapt to evolving threats, and use technology to drive growth and resilience. The success and sustainability of financial institutions will depend on how effectively they integrate innovative technologies with robust cybersecurity measures, ensuring they remain competitive and trustworthy in an interconnected world.

The Role of Social Media Algorithms in Modern Society

Social media platforms have become powerful gatekeepers of news, information, and opinion, shaping how citizens consume content. As such, they must recognize their significant responsibilities, including promoting the public good and preserving democracy. While platforms are free to use algorithms for legitimate goals, they should seek input from journalism professionals to ensure ethical standards guide their operations. Greater transparency is needed, especially regarding data collection and its potential impact on users' rights to seek, receive, and share information. Social media companies should reassess their business models to prioritize quality and accuracy over speed and quantity, ensuring that news distribution serves the public's information needs. Given their dominance in the market, social media platforms must embrace their responsibilities as news distributors and publishers. By placing a commitment to freedom of expression and respect for informational rights above profit-driven goals, they can help safeguard democracy and promote ethical practices in the digital age.

Advances in Computer Hardware (2015–2024)

NVIDIA, ASUS, and Libre are key players in the SBC market, with NVIDIA leading due to its advanced GPU technologies. Most SBCs, except for newer models in the Jetson series, are more suited for entry-level computer vision (CV) tasks due to limitations in memory, GPU performance, and cooling. Newer Jetson models, like the Orin Nano and Orin AGX, offer powerful GPU architectures but come at a higher price. Challenges in deploying CV models on SBCs include limited resources and real-time processing. Optimization techniques like quantization, pruning, and federated learning are explored to overcome these. Libraries such as TensorFlow and PyTorch are popular, with PyTorch offering flexibility. Key frameworks include Detectron2, TAO Toolkit, and OpenMMLab, with TensorRT providing optimal performance on NVIDIA boards. However, SBCs face limitations in power, memory, real-time processing, and heat management. Future research should focus on better hardware designs and optimized algorithms for SBCs in CV applications.

Computing and Environmental Sustainability

Global warming and environmental concerns have drawn attention worldwide, especially with the increasing energy usage in data centers, which significantly impacts the environment through CO₂ emissions. Green computing has emerged as a solution, encouraging the integration of ecological sustainability into ICT design, production, and disposal, aiming to reduce emissions, waste, and energy consumption. This approach fosters environmental conservation while improving efficiency and promoting green economic growth. Virtualization, a key technology in green computing, reduces energy usage in data centers by optimizing resource allocation and minimizing redundant infrastructure. This technology not only decreases operating costs but also significantly reduces greenhouse gas emissions, contributing to environmental sustainability. IT infrastructures currently contribute approximately 2% of global CO₂ emissions, and without energy-efficient practices, this figure is expected to grow. Therefore, adopting energy-efficient techniques and algorithms is crucial to managing computing resources sustainably.

The Development of Virtual Reality (VR) and Augmented Reality (AR)

Augmented Reality (AR) and Virtual Reality (VR) are transformative technologies with the potential to redefine education by offering immersive and engaging learning experiences. These tools enable students to interact with visually rich environments, audio cues, and simulations, enhancing their interest and understanding of various subjects. AR and VR also create engaging virtual worlds, making the learning process more dynamic and interactive. From 2011 to 2022, AR and VR have increasingly been explored in education, focusing on their benefits, challenges, and applications. These technologies saw exponential growth during the COVID-19 pandemic as remote learning emphasized the need for innovative educational tools. Despite their integration being in its early stages, AR and VR have shown the capacity to reshape traditional education by offering new ways to teach and learn, benefiting students, educators, and institutions alike.

Trends in Computing Education (2015–2024)

Today, continuous learning is a necessity due to the rapid change in technology and the environment in which we live. Moreover, traditional knowledge acquisition in academia settings is not sufficient anymore, the unpredictable demands of the future forcing one to focus also on critical thinking and adaptability. In professional settings, ongoing learning is essential for achieving success in one's career. Moreover, a valuable competence would be to continuously update the skills to become proficient in navigating the complexities of ever-changing worlds. Between 2015 and 2024, computing education has undergone significant transformations, driven by technological advancements and evolving educational methodologies. The integration of cloud computing has facilitated scalable and cost-effective educational solutions, enhancing accessibility and collaboration among students and educators. The proliferation of Artificial Intelligence (AI) has prompted educational institutions to update curricula and introduce new programs to equip students with AI skills, reflecting the growing demand for these competencies in the workforce. Extended Reality (XR) technologies, including Augmented Reality (AR) and

Virtual Reality (VR), have become more mainstream, offering immersive and personalized learning experiences. Additionally, there has been a concerted effort to align computing curricula with current industry standards, incorporating contemporary technologies and trends such as cybersecurity, machine learning, and artificial intelligence, better preparing students for the rapidly evolving IT industry. In summary, the period from 2015 to 2024 has been characterized by a dynamic evolution in computing education, driven by technological advancements and the imperative to meet the changing demands of the industry.

ANALYSIS

The analysis of technological trends reveals several key themes, including the rapid evolution of Artificial Intelligence (AI), quantum computing, open-source software, and advancements in computing hardware. These innovations have been driven by increased computational power, interdisciplinary collaboration, and data-centric methodologies. AI has transitioned from symbolic methods to data-driven techniques, achieving significant milestones in fields such as machine learning, natural language processing (NLP), and robotics. Quantum computing, while promising transformative applications in areas like drug discovery and climate modeling, still faces challenges in scalability and error correction. Open-source software has gained widespread adoption, particularly in enterprise environments, due to its cost-effectiveness, flexibility, and capacity to foster innovation. Similarly, green computing emphasizes the need for sustainable practices in IT, reducing environmental impacts through energy-efficient algorithms and virtualization technologies. The societal and industrial impacts of these advancements are profound. AI has enhanced efficiency and decision-making in healthcare, finance, and gaming, while quantum computing offers solutions to previously unsolvable problems. Open-source software democratizes access to technology, enabling smaller enterprises to innovate alongside established players. However, these advancements are not without drawbacks. Ethical concerns such as AI bias, privacy risks, and the societal implications of algorithm-driven platforms demand greater oversight and transparency. Environmental challenges from resource-intensive technologies further emphasize the importance of sustainable practices, as seen in the adoption of green computing. Looking ahead, the integration of explainable AI, energy-efficient technologies, and extended reality (XR) tools is expected to shape future developments. Hybrid quantum-classical systems may see wider deployment, addressing complex problems with enhanced precision. The evolution of computing education will play a critical role, with curricula increasingly aligning with industry demands for AI, cybersecurity, and XR skills. Emerging technologies like decentralized systems and advanced algorithms are poised to redefine interactions in sectors such as education, healthcare, and finance. As these trends unfold, the emphasis on ethical considerations, sustainability, and global collaboration will be essential to ensure these technologies serve humanity responsibly and equitably. This comprehensive outlook underscores the need for continued research and thoughtful integration of these innovations into society.

DISCUSSION

Reflecting on the analysis, it becomes evident that the evolution of computing has been marked by transformative advancements that intertwine innovation, ethics, and societal impact. Each technological trend discussed demonstrates a profound shift not only in computational capabilities but also in the way technology integrates into daily life and industry. AI, for instance, has transitioned from theoretical exploration to practical applications that are now ubiquitous across healthcare, finance, and education. Similarly, quantum computing, while still in its nascent stages, underscores humanity's pursuit of solving problems once deemed insurmountable, pushing the boundaries of what is computationally possible. Open-source software and green computing highlight collaborative and sustainable approaches, reflecting the industry's pivot towards accessibility and environmental responsibility. The findings contribute significantly to understanding the evolution of computing by emphasizing its multidimensional growth. Technological advancements are no longer isolated to performance improvements; they now incorporate ethical principles, global collaboration, and sustainability. This perspective broadens the understanding of computing beyond technical metrics to include its impact on society, governance, and the environment. The analysis also underscores the iterative nature of innovation, where breakthroughs in one area often spur developments in others. For instance, AI benefits from open-source tools, while green computing technologies influence hardware and infrastructure designs for both AI and quantum computing. The relevance of these advancements to current and future industry practices is undeniable. In today's landscape, industries must adopt technologies like AI and blockchain not just to remain competitive but to address challenges such as security, scalability, and efficiency. For instance, AI-driven automation is reshaping operational workflows, while decentralized technologies ensure data integrity in sensitive sectors. Looking ahead, the role of green computing and energy-efficient infrastructure will be critical as industries face increasing regulatory and societal pressure to reduce their environmental footprint. Furthermore, the ongoing integration of extended reality (XR) into training and education signals a shift towards more immersive and adaptable learning environments, equipping the workforce with the skills required for a rapidly changing technological landscape. In conclusion, the findings highlight a trajectory of computing that is as much about addressing global challenges as it is about technological progress. By embracing ethical standards, fostering collaboration, and prioritizing sustainability, the industry is well-positioned to leverage these advancements for meaningful and equitable growth. This discussion underscores the importance of aligning technological evolution with societal and environmental considerations to ensure a balanced and impactful future.

CONCLUSION

The analysis of technological trends and their implications offers several key insights into the evolution of computing over the past decade. Artificial Intelligence (AI) has emerged as a cornerstone of modern technology, driving innovation across industries through machine learning, natural language processing, and robotics. Quantum computing has shown immense potential for solving complex problems, though scalability and practical implementation remain hurdles. Open-source software has democratized technology, fostering collaboration and cost-efficiency, while green computing emphasizes the importance of sustainability in IT practices. Furthermore, the integration of extended reality (XR) into education and industry highlights the growing role of immersive technologies in shaping the future workforce. These advancements collectively reflect a shift towards more accessible, sustainable, and ethical computing practices, emphasizing their multidimensional growth. The impact of these advancements is transformative, reshaping industries and society in profound ways. AI-driven automation and decision-making tools have enhanced productivity and innovation in sectors like healthcare, finance, and education. Quantum computing promises breakthroughs in drug discovery, logistics, and climate modeling, despite its developmental challenges. The rise of open-source frameworks and platforms has lowered barriers to entry for startups and small enterprises, fueling innovation and competition. At the same time, the prioritization of environmental sustainability through green computing demonstrates a growing awareness of technology's ecological footprint. These technological milestones underscore the importance of aligning innovation with ethical principles, transparency, and sustainability to ensure a balanced impact. Looking to the future, several areas warrant further exploration and research. The ethical challenges associated with AI, such as bias, transparency, and accountability, require ongoing attention to ensure fair and equitable technology deployment. Quantum computing research should focus on achieving practical scalability and error correction, enabling real-world applications across industries. Additionally, the development of more energy-efficient hardware and algorithms in line with green computing principles is crucial as data centers and IT infrastructures expand. Finally, exploring the societal implications of emerging technologies like XR and blockchain will help shape policies and frameworks that prioritize inclusivity and security. In summary, the past decade's technological advancements have profoundly influenced the trajectory of computing, providing valuable lessons about the importance of innovation, ethics, and sustainability. As we move forward, the challenge lies in balancing rapid technological progress with the need for thoughtful integration, ensuring these advancements

contribute meaningfully to societal and environmental well-being. Future research should focus on addressing unresolved challenges while exploring opportunities for interdisciplinary collaboration and global impact.

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